

Deconstructing Multiple Correspondence Analysis



Jan de Leeuw

1 Notation

Let us start by defining some of the notation used in this paper. We have $i = 1, \dots, n$ observations on each of $j = 1, \dots, m$ categorical variables, where variable j has k_j categories. We use k_* for the sum of the k_j , while the maximum number of categories over all variables is $k_+ = \max(k_1, \dots, k_m)$. We also define m_s , with $s = 1, \dots, k_+$, where m_s is the number of variables with $k_j \geq s$. Thus both m_1 and m_2 are always equal to m . Also $\sum_{s=1}^{k_+} m_s = k_*$. The fact that variables can have a different number of categories is a major notational nuisance. If they all have the same number of categories k then $k_+ = k$, $k_* = mk$, and all m_s are equal to m .

The data are coded as m indicator matrices $\mathbf{G}_j$, with $\{\mathbf{G}_j\}_{ik} = 1$ if and only if object i is in category k of variable j and $\{\mathbf{G}_j\}_{ik} = 0$ otherwise. The $\mathbf{G}_j$ are $n \times k_j$, zero-one, and column-wise orthogonal (because the categories are mutually exclusive). If we concatenate the $\mathbf{G}_j$ horizontally we have the $n \times k_*$ matrix $\mathbf{G}$, which we also call the indicator matrix; in French data analysis it is the “tableau disjonctif complet”, in Nishisato (1980) it is the “response-pattern table”. The Burt table (“tableau de Burt”), is the $k_* \times k_*$ cross product matrix $\mathbf{C} = n^{-1} \mathbf{G}' \mathbf{G}$. The univariate marginals are in the diagonal matrix $\mathbf{D} = \text{diag}(\mathbf{C})$. The normalised Burt table is the matrix $\mathbf{E} = m^{\frac{1}{2}} \mathbf{D}^{-\frac{1}{2}} \mathbf{C} \mathbf{D}^{-\frac{1}{2}}$.

Although we introduced $\mathbf{G}$, $\mathbf{C}$, $\mathbf{D}$ and $\mathbf{E}$ as partitioned matrices of real numbers, it is also useful to think of them as matrices with matrices as elements. Thus $\mathbf{C}$, for example, is an $m \times m$ matrix with as elements the matrices $\mathbf{C}_{j\ell} = \mathbf{G}'_j \mathbf{G}_\ell$, and $\mathbf{G}$ is an $1 \times m$ matrix with as its m elements $\mathbf{G}_j$. Note that because we have divided the cross product by n , all $\mathbf{C}_{j\ell}$, and thus all $\mathbf{D}_j = \mathbf{C}_{jj}$, add up to one.

J. de Leeuw (✉)

Department of Statistics, University of California Los Angeles (UCLA), Los Angeles, CA, USA
e-mail: deleeuw@stat.ucla.edu

In the paper we often use the direct sum of matrices. If $\mathbf{A}$ and $\mathbf{B}$ are matrices, then their direct sum is:

$$\mathbf{A} \oplus \mathbf{B} = \begin{bmatrix} \mathbf{A} & \mathbf{0} \\ \mathbf{0} & \mathbf{B} \end{bmatrix}, \quad (1)$$

and if $\mathbf{A}_r$ are s matrices, then $\bigoplus_{r=1}^s \mathbf{A}_r$ is block-diagonal with the $\mathbf{A}_r$ as diagonal submatrices.

2 Introduction

Multiple Correspondence Analysis (MCA) can be introduced in many different ways.

Mathematically: MCA is the Singular Value Decomposition (SVD) of $m^{-\frac{1}{2}}\mathbf{G}\mathbf{y} = \sqrt{\lambda}\mathbf{x}$ and $m^{-\frac{1}{2}}\mathbf{G}'\mathbf{x} = \sqrt{\lambda}\mathbf{D}\mathbf{y}$, the Eigen Value Decomposition (EVD) $\mathbf{E}\mathbf{y} = \lambda^2\mathbf{y}$ for the normalised Burt table, and the EVD of $m^{-1}\mathbf{G}'\mathbf{D}^{-1}\mathbf{G}\mathbf{x} = \lambda^2\mathbf{x}$, the average projector. Using m in the equations seems superfluous, but it guarantees that $0 \leq \lambda \leq 1$.

Statistically: MCA is a scoring method that minimises the within-individual and maximises the between-individuals variance, it is a graphical biplot method that minimises the distances between individuals and the categories of the variables they score in, it is an optimal scaling method that maximises the largest eigenvalue of the correlation matrix of the transformed variables, and that linearises the average regression of one variable with all the others. It can also be presented as a special case of Homogeneity Analysis, Correspondence Analysis, and Generalised Canonical Correlation analysis; see, for example, the review article by Tenenhaus and Young (1985).

It is of some interest to trace the origins of these various MCA formulations, and to relate them to an interesting exchange in the 1950s between two of the giants of psychometrics on whose proverbial shoulders we still stand. In 1950 Sir Cyril Burt published, in his very own British Journal of Statistical Psychology, a great article introducing MCA as a form of factor analysis of qualitative data (Burt 1950). There are no references in the paper to earlier occurrences of MCA in the literature. This prompted Louis Guttman to point out in a subsequent issue of the same journal that the relevant equations were already presented in great detail in Guttman (1941). Guttman assumed Burt had not seen the monograph (Horst 1941) in which his chapter was published, because of the communication problems during the war, which caused “only a handful of copies to reach Europe” (Guttman 1953). Although the equations and computations given by both Burt and Guttman were identical, Guttman pointed out differences in interpretation between their two approaches. These differences will especially interest us in the present paper. They were also discussed in Burt’s reaction to Guttman’s note (Burt 1953). The three papers are still very readable and instructive, and in the first part of the present paper we’ll put them in an historical context.

3 History

3.1 Prehistory

The history of MCA has been reviewed in De Leeuw (1973), Benzécri (1977b), Nishisato (1980, Sect. 1.2), Tenenhaus and Young (1985), Gower (1990), and Lebart and Saporta (2014), each from their own tradition and point of view. Although there is agreement on the most important stages in the development of the technique, there are some omissions and some ambiguities. Some of the MCA historians, in their eagerness to produce a long and impressive list of references, do not seem to distinguish multiple from ordinary Correspondence Analysis (CA), one-dimensional from multidimensional analysis, binary data from multicategory data, and data with or without a dependent variable.

What we call “prehistory” is MCA before Guttman (1941), and what we find in the prehistory is almost exclusively Reciprocal Averaging Analysis (RAA). We define RAA, in the present paper, starting from the indicator matrix G . Take any set of trial weights for the categories. Then compute the score for the individual by averaging the weights of the categories selected by that individual, and then compute a new set of weights for categories by averaging the scores of the individuals in the categories. These two reciprocal averaging steps are iterated until convergence is attained, that is when weights and scores do not change any more (up to a proportionality factor).

In various places it is stated, or at least suggested, that RAA (both the name and the technique) started with Richardson and Kuder (1933). This seems incorrect. That paper has no trace of RAA, although it does document a scale construction using Hollerith sorting and tabulation machines. What seems to be true, however, is that both the RAA name and the technique started at Proctor and Gamble in the early 1930s, in an interplay between Richardson and Horst, both Proctor and Gamble employees at the time. This relies on the testimony of Horst (1935), who does indeed attribute the name and basic idea of RAA to Richardson:

The method which he suggested was based on the hypothesis that the scale value of each statement should be taken as a function of the average score of the men for whom the statement was checked and, further, that the score of each man should be taken as a function of the average scale value of the statements checked for the man.

The definition given by Horst is rather vague, because “a function of” is not very specific. It also does even mention the iteration of RAA to convergence (or, as Guttman would say, internal consistency). This iterative extension again seems to be due either to Horst or to Richardson. Horst was certainly involved at the time in the development of very similar techniques for quantitative data (Horst 1936; Edgerton and Kolbe 1936; Wilks 1938). For both quantitative and qualitative data these techniques are based on minimising within-person and maximising between-person variance, and they all result in computing the leading principal component of some data matrix. Horst (1935), starting from the idea to make linear combinations to maximise between individual variance, seems to have been the first one to realise

that the equations defining RAA are the same as the equations describing Principal Component Analysis (PCA), and that consequently there are multiple RAA solutions for a given data matrix.

There are some additional hints about the history of RAA in the conference paper of Baker and Hoyt (1972). They also mostly credit Richardson, although they mention he never published a precise description of the technique, and it has been used “informally” without a precise justification ever since. They also mention that the first Hollerith type of computer implementation of RAA was by Mosier in 1942, the first UNIVAC programme was by Baker in 1962, and the first FORTRAN programme was by Baker and Martin in 1969.

We have not mentioned in our prehistory the work of Fisher (1938, 1940) and Maung (1941). These contributions, basically contemporaneous with Guttman (1941), clearly introduced the idea of optimal scaling for categorical data, of Correspondence Analysis of a two-way table, and even of non-linear transformation of the data to fit a linear (additive) model. They also came up with the first principal component of a Gramian matrix as a solution, realising there are multiple solutions to their equations. However, as pointed out by Gower (1990), they do not use MCA as it is currently defined. And, finally, although Hill (1973) seems to have independently come up with the RAA name and technique, its origins are definitely not in ecology.

3.2 *Guttman 1941*

RAA was used to construct a single one-dimensional scale, but Horst (1935) indicated already its extension to more than one dimension. The first publication of the actual formulas, using the luxuries of modern matrix algebra, was Guttman (1941), ironically in a chapter of a book edited by Horst. This is really where the history of MCA begins, although there are still some notable differences with later practise.

Guttman starts with the indicator matrix $\mathbf{G}$, and then generalises and systematises the analysis of variance approach to optimal scaling of indicator matrices. He introduces three criteria of internal consistency: one for the categories (columns), one for the objects (rows), and one for the entire table. All three criteria lead to the same optimal solution, which we now recognise as the first non-trivial dimension of MCA. We now also know, because we have been exposed to more matrix algebra than was common in the 1940s and 1950s, that this merely restates the fact that for any matrix $\mathbf{X}$ the non-zero eigenvalues of $\mathbf{X}'\mathbf{X}$ and $\mathbf{X}\mathbf{X}'$ are the same, and moreover they are equal to the squares of the singular values of $\mathbf{X}$. The left and right singular vectors of $\mathbf{X}$ are the eigenvectors of $\mathbf{X}'\mathbf{X}$ and $\mathbf{X}\mathbf{X}'$.

For our purposes in this paper the following quotation from Guttman’s section five is important. When discussing the multiple solutions of the MCA stationary equations he says (pp. 330–331):

There is an essential difference, however, between the present problem of quantifying a class of attributes and the problem of “factoring” a set of quantitative variates. The principal axis solution for a set of quantitative variates depends on the preliminary units of measurement

of those variates. In the present problem, the question of preliminary units does not arise since we limit ourselves to considering the presence or absence of behaviour.

Thus Guttman, at least in 1941, shows a certain reluctance to consider the additional dimensions in MCA for data analysis purposes.

In addition to the stationary equations of MCA, Guttman also introduces the chi-square metric. He notes that the rank of $\mathbf{C}$, and thus of $\mathbf{E}$, is that of the indicator $\mathbf{G}$, which is at most $1 + \sum_{j=1}^m (k_j - 1) = k_* - (m - 1)$. Thus $\mathbf{C}$ has at least $m - 1$ zero eigenvalues, inherited from the linear dependencies in $\mathbf{G}$. In addition $\mathbf{E}$ has a trivial eigen pair, independent of the data, with eigenvalue equal to 1. Suppose the vector $\mathbf{e}$ has all its k_* elements equal to +1. Then $\mathbf{C}\mathbf{e} = m\mathbf{D}\mathbf{e}$ and thus $\mathbf{E}\mathbf{y} = \mathbf{y}$, with $\mathbf{y} = \mathbf{D}^{\frac{1}{2}}\mathbf{e}$. If we deflate the eigenvalue problem by removing this trivial solution then the sum of squares of any off-diagonal submatrix of $\mathbf{C}$ is the chi-square for independence of that table.

Guttman also points out that the scores and weights linearise both regressions if we interpret the indicator matrix as a discrete bivariate distribution. This follows directly from the interpretation of MCA as a CA of the indicator matrix, because CA linearises regressions in a bivariate table. Of course interpreting the binary indicator matrix $\mathbf{G}$ as a bivariate distribution is quite a stretch. Both the chi-square metric and the linearised regressions were discussed earlier by Hirschfeld (1935) in the context of a single bivariate table. Neither Hirschfeld nor Fisher are mentioned in Guttman (1941).

There are no data and examples in Guttman's article. Benzécri (1977b) remarks:

L. Guttman avait défini les facteurs mêmes calculés par l'analyse des correspondances. Il ne les avait toutefois pas calculés; pour la seule raison qu'en 1941 les moyens de calcul requis (ordinateurs) n'existaient pas.

Translation: L. Guttman has defined the same factors as calculated by Correspondence Analysis. He did not calculate them, however, for the simple reason that in 1941 the necessary calculation tools (computers) did not exist.

That is not exactly true. In Horst (1941) the chapter by Guttman is followed by another chapter called "Two Empirical Studies of Weighting Techniques", which does have an empirical application in it. It is unclear who wrote that chapter, but the computations, which were carried out on a combination of tabulating and calculating machines, were programmed by nobody less than Ledyard R. Tucker.

3.3 Burt 1950

Guttman was reluctant to look at additional solutions of the stationary equations (additional "dimensions"), but Burt (1950) had no such qualms. After a discussion of the indicator matrix $\mathbf{G}$ and its corresponding cross product $\mathbf{C}$ (now known as the Burt table) Burt suggests a PCA of the normalised Burt table, i.e. solving the eigen problem $\mathbf{E}\mathbf{y} = m\lambda\mathbf{y}$. By the way, Burt discusses PCA as an alternative method of factor analysis, which is not in line with current usage clearly distinguishing PCA and FA.

Most of Burt's references are to previous PCA work with quantitative variables, and much of the paper tries to justify the application of PCA to qualitative data. No references to Guttman, Fisher, Horst, or Hirschfeld are given. The justifications that Burt presents are from the factor analysis perspective: $\mathbf{C}$ is a Gramian matrix, $\mathbf{E}$ is a correlation matrix, and the results of factoring $\mathbf{E}$ can lead to useful classifications of the individuals.

In the technical part of Burt's 1950 paper he discusses the rank, the trivial solutions, and the connection with the chi-squares of the bivariate subtables that we have already mentioned in our "Guttman (1941)" section.

3.4 *Guttman 1953*

As we saw in the introduction Guttman (1953) starts his paper with the observation that he already published the MCA equations in 1941. He gives this a positive spin, however, stating (p. 1):

It is gratifying to see how Professor Burt has independently arrived at much the same formulation. This convergence of thinking lends credence to the suitability of the approach.

I will now insert a long quote from Guttman (1953, p. 2), because it emphasises the difference with Burt, and it is of major relevance to the present paper as well. Guttman really tells it like it is:

My own article goes on to point out that, while the principal components here are formally similar to those for quantitative variables, nevertheless their interpretation may be quite different. The interrelations among qualitative items are not linear, nor even algebraic, in general. Similarly, the relation of a qualitative item to a quantitative variable is in general non-algebraic. Since the purpose of principal components—or any other method of factor analysis—is to help reproduce the original data, one must take into account this peculiar feature.

The first principal component can possibly fully reproduce all the qualitative items entirely by itself: the items may be perfect, albeit non-algebraic, functions of this component. *Linear* prediction will not be perfect in this case, but this is not the best prediction technique possible for such data. Therefore, if the first principal component only accounts for a small proportion of the total variance of the data in the ordinary sense, it must be remembered that this ordinary sense implies linear prediction. If the correct, but *non-linear*, prediction technique is used, the whole variation can sometimes be accounted for by but the single component. In such a case, the existence of more than one principal component arises merely from the fact that a linear system is being used to approximate a non-linear one. (Each item is always a perfect linear function of *all* the principal components taken simultaneously).

This was written after the publication of Guttman (1950) in which the MCA of a perfect scale of binary items is discussed in impressive detail. The additional dimensions in such an analysis are curvilinear functions of the first, in regular cases in fact orthogonal polynomials of a single scale. Specifically, the second dimension is a quadratic, or quadratic-looking, function of the first, which creates the famous "horseshoe" or "arch" (in French: the "effect Guttman"). Since a horseshoe curves

back in at its endpoints that name is often not appropriate, and we will call these non-linearities the Guttman effect. It seems that the second and higher curved dimensions are just mathematical artefacts, and much has been published since 1950 to explain them, interpret them, or to get rid of them (Hill and Gauch 1980).

In the rest of Guttman (1953) gives an overview of more of his subsequent work on scaling qualitative variables. This leads to material that goes beyond MCA (and thus beyond the scope of our present paper).

3.5 *Burt 1953*

Burt (1953, p. 5), in his reply to Guttman (1953), admits there are different objectives involved:

If, as I gather, he cannot wholly accept my own interpretations, that perhaps is attributable to the fact that our starting-points were rather different. My aim was to factorise such data; his to construct a scale.

This does not answer the question, of course, if it is really advisable to apply PCA to the normalised Burt matrix. It also seems there also are some differences in national folklore, since Burt (1953, p. 6) goes on to say:

In the chapters contributed to **Measurement and Prediction** both Dr. Guttman and Dr. Lazarsfeld draw a sharp distinction between the principles involved in these two cases. Factor analysis, they maintain, has been elaborated solely with reference to data which is quantitative **ab initio**; hence, they suppose, it cannot be suitably applied to qualitative data. On this side of the Atlantic, however, there has always been a tendency to treat the two cases together, and, with this double application in view, to define the relevant functions in such a way that they will (so far as possible) cover both simultaneously. British factorists, without specifying very precisely the assumptions involved, have used much the same procedures for either type of material. Nevertheless, there must of necessity be certain minor differences in the detailed treatment. These were briefly indicated in the paper Dr. Guttman has cited; but they evidently call for a closer examination. I think in the end it will be found that they are much slighter than might be supposed.

Burt then goes on to treat the case of a perfect scale of binary items, previously analysed by Guttman (1950). He points out that a PCA of a perfect scale gives (almost) the same results as those given by Guttman, and that consequently his approach of factoring a table works equally well as the approach that constructs a scale. Indeed, the differences between qualitative and quantitative factoring are “much slighter than might be supposed”. Although Burt is correct, he does not discuss where the Guttman effect comes from, and whether it is desirable and/or useful.

3.6 *Benzécri 1977*

French data analysis (“Analyse des Données”) views MCA as a special case of CA (Le Roux and Rouanet 2010). Benzécri (1977a) discusses the CA of the indicator matrix and gives a great deal of credit to Ludovic Lebart. Lebart (1975, 1976) are usually mentioned as the first publications to actually use “analyse de correspondances multiples” and “tableau de Burt”.

Benzécri also gives Lebart the credit for discovering that a CA of the indicator matrix $\mathbf{G}$ gives the same results as a CA of the Burt table $\mathbf{C}$, which restates again our familiar matrix result that the singular value decomposition of a matrix gives the same results as the eigen decomposition of the two corresponding cross product matrices:

L. Lebart en apporta la meilleure justification: les facteurs sur J issus de l’analyse d’un tel tableau $I \times J$ ne sont autres (à un coefficient constant près) que ceux issus de l’analyse du véritable tableau de contingence $J \times J$ suivant: $k(j, j') =$ nombre des individus i ayant à la fois la modalité j et la modalité j' . Dès lors on rejoint le format original pour lequel a été conçue l’analyse des correspondances.

Translation: L. Lebart has given the best justification: the factors on J from an analysis of an $I \times J$ table are the same as those from the analysis of the actual $J \times J$ contingency table with $k(j, j') =$ the number of individuals i that are both in category j and j' . And thus we are back in the original format for which Correspondence Analysis was designed.

Benzécri also mentions the surprising generality and wide applicability of MCA:

Le succès maintenant bien compris des analyses de tableaux en 0,1 mis sous forme disjonctive complète invite à rapprocher de cette forme, par un codage approprié, les données les plus diverses.

Translation: The success, which we now understand well, of the analysis of (0,1) tables in disjunctive complete form invites us to apply this form, by suitable coding, to the most diverse forms of data.

This generality was later fully exploited in the book by Gifi (1990), which builds a whole system of descriptive multivariate techniques on top of MCA.

3.7 *Gifi 1980*

Gifi (1990) was mostly written in 1980–1981 from lecture notes for a graduate course in non-linear multivariate analysis, and builds on previous work in De Leeuw (1973). Throughout, the main engine of the Gifi approach to multivariate analysis minimises the meet-loss function:

$$\sigma(\mathbf{X}; \mathbf{Y}_1, \dots, \mathbf{Y}_m) = \sum_{j=1}^m \text{tr}(\mathbf{X} - \mathbf{G}_j \mathbf{Y}_j)' (\mathbf{X} - \mathbf{G}_j \mathbf{Y}_j), \quad (2)$$

over the $n \times p$ matrices of scores $\mathbf{X}$ with $\mathbf{X}'\mathbf{X} = n\mathbf{I}$ and over the $k_j \times p$ matrices of loadings $\mathbf{Y}_j$ that may or may not satisfy some constraints. Gifi calls this general approach Homogeneity Analysis (HA). Loss function (2) was partly inspired by Carroll (1968) who used this least squares loss function in generalised canonical analysis of quantitative variables.

The different forms of multivariate analysis in the Gifi framework arise by imposing additivity, and/or rank, and/or ordinal constraints on the $\mathbf{Y}_j$. See De Leeuw and Mair (2009) for a user's guide to the R package `homals`, which implements minimisation of meet-loss under these various sets of constraints.

If there are no constraints on the $\mathbf{Y}_j$ then minimising (2) computes the p dominant dimensions of an MCA. What makes the loss function (2) interesting in our comparative review of MCA is the distance interpretation and the corresponding geometry of the joint biplot of objects and categories. Gifi minimises the sum of the squared distances between an object and the categories of the variables that the object scores are in. If we make a separate biplot for each variable j it has n object points and k_j category points. The category points are in the centroid of the object points in that category, and if we connect all those objects with their category points we get k_j star graphs in what Gifi calls the star plot. Minimising (2) means making the joint plot in such a way that the stars are as small as possible.

The `homals` package of De Leeuw and Mair (2009) actually computes the proportion of individuals correctly classified if we assign each individual to the category it is closest to (in p dimensions). In this way we can indeed find, like Guttman, that a single component can account for all of the "variance".

There are indications, especially in Gifi (1990, Sect. 3.9), that they are somewhat uncomfortable with the multidimensional scale construction aspects of MCA. They argue that each MCA dimension gives a quantification or transformation of the variables, and thus each MCA dimension can be used to compute a different correlation matrix between the variables. These correlation matrices, of which there are $k_\star - m$, can then all be subjected to a PCA. So the single indicator matrix leads to $k_\star - m$ PCA's. Gifi calls this "data production", and obviously does not like the outcome. Thus, as an alternative to MCA, they suggest using only the first dimension and the corresponding correlation matrix, which is very close to RAA and to Guttman (1941).

In the Gifi system the data production dilemma is further addressed in two ways. In the geometric framework based on the loss function (2) a form of non-linear PCA is defined in which we restrict the $k_j \times p$ category quantifications of a variable to have rank one, i.e. the points representing the categories of a variable are on a line through the origin. Gifi shows that this leads to the usual non-linear PCA techniques (De Leeuw 2006; Young et al. 1978). The second development to get away from the "data production" in MCA is the "aspect" approach (De Leeuw 1988a,b, 2004; Mair and De Leeuw 2010; De Leeuw et al. 1999). There we look for a single quantification or transformation of the variables that optimises any real valued function (or aspect) of the resulting correlation matrix. Non-linear PCA is the special cases in which we maximise the sum of the first p eigenvalues of the correlation matrix, and MCA chooses the scale to maximise the dominant eigenvalue. Other aspects

lead to regression, canonical analysis, and structural equation models. In this more recent methodology based on aspects Guttman's one-dimensional scale construction approach has won out over Burt's multidimensional factoring method.

4 Deconstructing MCA

4.1 Introduction

We are left with the following questions from our history section, and from the Burt-Guttman exchange:

1. What, if anything, is the use of additional dimensions in MCA?
2. Where does the Guttman effect come from?
3. Is MCA really just PCA?
4. How many dimensions of MCA should we keep?
5. Which "variance" is "explained" by MCA?
6. How do we handle the "data production" aspects of MCA?

In De Leeuw (1982) several results are discussed that are of importance in answering these questions, and more generally for the interpretation (and deconstruction) of MCA. Additional, and more extensive, discussion of these same results is in Bekker and De Leeuw (1988) and De Leeuw (1988a).

To compute the MCA eigen decomposition we could, for example, use the Jacobi method, which diagonalises $\mathbf{E}$ by using elementary plane rotations. It builds up $\mathbf{Y}$ by minimising the sum of squares of the off-diagonal elements. Thus $\mathbf{E}$ is updated by iteratively replacing it by $J_{st}\mathbf{E}J_{st}$, where J_{st} with $s < t$ is a Jacobi rotation, that is, a matrix that differs from the identity matrix of order k_* only in elements (s, s) and (t, t) , which are equal to u , and in elements (s, t) and (t, s) which are $+v$ and $-v$, where u and v are real numbers with $u^2 + v^2 = 1$. We cycle through all upper-diagonal elements $s < t$ for a single iteration, and continue iterating until the $\mathbf{E}$ update is diagonal (within some ϵ).

We shall discuss a different three-step method of *approximately* diagonalising $\mathbf{E}$, which, for lack of a better term, we call Deconstructed Multiple Correspondence Analysis (DMCA). It also works by applying elementary plane rotations to $\mathbf{E}$, but it is different from the Jacobi method because it is not intended to exactly diagonalise any arbitrary real symmetric matrix, or any normalised Burt matrix for that matter. It uses its rotations to eliminate all off-diagonal elements of all m^2 submatrices $\mathbf{E}_{kl}$, where $k, l = 1, \dots, m$. If it cannot do this perfectly, it will try to find the best approximate diagonalisation. If DMCA does exactly diagonalise all submatrices, then some rearranging and additional computation finds the eigenvalues and eigenvectors of $\mathbf{E}$, and thus the MCA. The eigenvectors are, however, ordered differently (not by decreasing eigenvalues), and provide more insight in the inner workings of MCA. If an exact diagonalisation is not possible, the approximate diagonalisation often still provides this insight.

We first discuss some theoretical cases in which DMCA leads to the MCA, and after that some empirical examples are described in which the diagonalisation is only approximate and DMCA and MCA differ. As you will hopefully see, both types of DMCA examples show us what MCA as a data analysis technique tries to do, and how the results help in answering the six questions given above, arising from the Burt-Guttman exchange.

4.2 Mathematical Examples

4.2.1 Binary Data

Let's start with the case of binary data, i.e. indicator matrices for which all k_j are equal to two. The normalised Burt table $\mathbf{E} = m^{-1}\mathbf{D}^{-\frac{1}{2}}\mathbf{C}\mathbf{D}^{-\frac{1}{2}}$ consists of $m \times m$ submatrices $\mathbf{E}_{j\ell}$ of dimension 2×2 . Suppose the marginals of variable j are p_{j0} and p_{j1} . For each j make the 2×2 table:

$$\mathbf{K}_j = \begin{bmatrix} +\sqrt{p_{j0}} & +\sqrt{p_{j1}} \\ +\sqrt{p_{j1}} & -\sqrt{p_{j0}} \end{bmatrix}, \quad (3)$$

and suppose $\mathbf{K}$ is the direct sum of the $\mathbf{K}_j$, i.e. the block-diagonal matrix with the $\mathbf{K}_j$ on the diagonal. Then $\mathbf{F} = \mathbf{K}'\mathbf{E}\mathbf{K}$ again has $m \times m$ submatrices of order two. For each $j, \ell = 1, \dots, m$ the matrix $\mathbf{F}_{j\ell} = \mathbf{K}_j'\mathbf{E}_{j\ell}\mathbf{K}_\ell$ is diagonal, with element (1, 1) equal to +1 and element (2, 2) equal to the point correlation (or phi-coefficient) between binary variables j and ℓ (and thus also equal to +1 if $j = \ell$).

This means we can permute rows and columns of $\mathbf{F}$ using a permutation matrix $\mathbf{P}$ such that $\mathbf{R} = \mathbf{P}'\mathbf{F}\mathbf{P}$ is the direct sum of two correlation matrices $\mathbf{R}_{11}$ and $\mathbf{R}_{22}$, both of order m . $\mathbf{R}_{11}$ has all elements equal to +1, $\mathbf{R}_{22}$ has its off-diagonal elements equal to the phi-coefficients. We collect the (1, 1) elements of all $\mathbf{F}_{j\ell}$, which are all +1, in $\mathbf{R}_{11}$ and the (2, 2) elements in $\mathbf{R}_{22}$. Suppose $\mathbf{L}_1$ and $\mathbf{L}_2$ are the normalised eigenvectors of $\mathbf{R}_{11}$ and $\mathbf{R}_{22}$, and $\mathbf{L}$ is their direct sum. Then $\mathbf{\Lambda} = m^{-1}\mathbf{L}\mathbf{R}\mathbf{L}$ is diagonal, with on the diagonal the eigenvalues of $\mathbf{E}$ and with $\mathbf{K}\mathbf{P}\mathbf{L}$ the normalised eigenvectors of $\mathbf{E}$. Thus the eigenvalues of $\mathbf{E}$ are those of $m^{-1}\mathbf{R}_{11}$, i.e. one 1 and $m - 1$ zeros, together with those of $m^{-1}\mathbf{R}_{22}$. This restates the well-known result, mentioned by both Guttman (1941) and Burt (1950), that an MCA of binary data reduces to a PCA of the phi-coefficients.

4.2.2 Correspondence Analysis

Now let us look at Correspondence Analysis, that is, MCA with $m = 2$. There is only one single off-diagonal $p \times q$ cross table $\mathbf{C}_{12}$ in the Burt matrix. Suppose without loss of generality that $p \geq q$. Define $\mathbf{K}$ as the direct sum of the left and right singular vectors of $\mathbf{E}_{12}$. Then:

$$\mathbf{F} = \mathbf{K}'\mathbf{E}\mathbf{K} = \begin{bmatrix} \mathbf{I} & \Psi \\ \Psi' & \mathbf{I} \end{bmatrix}, \quad (4)$$

where Ψ is the $p \times q$ diagonal matrix of singular values of $\mathbf{E}_{12}$, and:

$$\mathbf{R} = \mathbf{P}'\mathbf{F}\mathbf{P} = \left\{ \bigoplus_{s=1}^q \begin{bmatrix} 1 & \psi_s \\ \psi_s & 1 \end{bmatrix} \right\} \bigoplus \mathbf{I}, \quad (5)$$

where the identity matrix at the end of Eq. (5) is of order $p - q$.

Thus the eigenvalues of $\mathbf{E}$ are $\frac{1}{2}(1 + \psi_s)$ and $\frac{1}{2}(1 - \psi_s)$ for all s , and DMCA indeed diagonalises $\mathbf{E}$. The relation between the eigen decomposition of $\mathbf{E}$ and the singular value decomposition of $\mathbf{E}_{12}$ is a classical result in Correspondence Analysis Benzécri (1977a), and earlier already in canonical correlation analysis of two sets of variables (Hotelling 1936).

4.2.3 Multinormal Distribution

Suppose we want to apply MCA to an m -variate standard normal distribution with correlation matrix $\mathbf{R} = \{\rho_{k\ell}\}$. Not to a sample, mind you, but to the whole distribution. This means we have to think of the submatrices $\mathbf{C}_{j\ell}$ as bivariate standard normal densities, having an infinite number of categories, one for each real number. Just imagine it as a limit of the discrete case (Naouri 1970).

In this case the columns of the $\mathbf{K}_j$, of which there is a denumerably infinite number, are the Hermite-Chebyshev polynomials $h_0, h_1, \dots$ on the real line. We know that for the standard bivariate normal $E_{j\ell}(h_s, h_t) = 0$ if $s \neq t$ and $E_{j\ell}(h_s, h_s) = \rho_{j\ell}^s$. Thus $\mathbf{F} = \mathbf{K}'\mathbf{E}\mathbf{K}$ is an $m \times m$ matrix of diagonal matrices, where each $\mathbf{F}_{k\ell}$ submatrix is of denumerably infinite order and has all the powers of $\rho_{k\ell}$ along the diagonal. Then $\mathbf{R} = \mathbf{P}'\mathbf{F}\mathbf{P}$ is the infinite direct sum of elementwise powers of the matrix of correlation coefficients, or:

$$\mathbf{R} = \mathbf{P}'\mathbf{F}\mathbf{P} = \bigoplus_{s=0}^{\infty} \mathbf{R}^{(s)}, \quad (6)$$

and $\mathbf{\Lambda} = \mathbf{L}'\mathbf{R}\mathbf{L}$ is diagonal, with the first m eigenvalues of $\mathbf{R}^{(0)} = \mathbf{e}\mathbf{e}'$, then the m eigenvalues of $\mathbf{R}^{(1)} = \mathbf{R}$, then the m eigenvalues of $\mathbf{R}^{(2)} = \{\rho_{j\ell}^2\}$, and so on to $\mathbf{R}^{(\infty)} = \mathbf{I}$. Each MCA solution is composed of Hermite-Chebyshev polynomials of the same degree. Again, this restates a known result, already given in De Leeuw (1973).

These results remain true for what Yule called “strained multinormals”, i.e. multivariate distributions that can be obtained from the multivariate normal by separate and generally distinct smooth monotone transformations of each of the variables. It also applies to mixtures of multivariate standard normal distributions with different

correlation matrices (Sarmanov and Bratoeva 1967), to Gaussian copulas, as well as to other multivariate distributions whose bivariate marginals have diagonal expansions in systems of orthonormal functions (the so-called Lancaster probabilities, after Lancaster (1958, 1969)).

The multinormal is a perfect example of the Guttman effect, that is, the eigenvector corresponding with the second largest eigenvalue usually is a quadratic function of the first, the next eigenvector usually is a cubic, and so on. We say “usually”, because Gifi (1990, pp. 382–384) gives a multinormal example in which the first two eigenvectors of an MCA are both linear transformations of the underlying scale (i.e. they both come from $\mathbf{R}_{22}$). However, the Guttman effect is observed approximately in many (if not most) empirical applications of MCA, especially if the categories of the variables have some natural order and if the number of individuals is large enough.

4.2.4 Common Mathematical Structure

What do our three previous examples have in common mathematically? In all three cases there exist orthonormal $\mathbf{K}_j$ and diagonal $\Phi_{j\ell}$ such that $\mathbf{E}_{j\ell} = \mathbf{K}_j \Phi_{j\ell} \mathbf{K}'_\ell$. Or, in words, the matrices $\mathbf{E}_{j\ell}$ in the same row-block of $\mathbf{E}$ have their left singular vectors $\mathbf{K}_j$ in common, and matrices $\mathbf{E}_{j\ell}$ in the same column-block of $\mathbf{E}$ have their right singular vectors $\mathbf{K}_\ell$ in common. Equivalently, this requires that for each j the m matrices $\mathbf{E}_{j\ell} \mathbf{E}_{\ell j}$ commute.

Another way of saying this is that there are vectors $\mathbf{y}_1, \dots, \mathbf{y}_m$ so that $\mathbf{C}_{j\ell} \mathbf{y}_\ell = \rho_{j\ell} \mathbf{D}_{jj} \mathbf{y}_j$, i.e. so that all bivariate regressions are linear (De Leeuw 1988a). Not only that, we assume that such a set of weights exist for every dimension s , as long as $k_j \geq s$. If $k_j = 2$ then trivially all regressions are linear, because you can always draw a straight line through two points. If $m = 2$ all Correspondence Analysis solutions linearise the regressions in a bivariate table. In the multinormal example the Hermite polynomials provide the linear regressions. Simultaneous linearisability of all bivariate regressions seems like a strong condition, which will never be satisfied for observed Burt matrices. But our empirical examples, analysed below, suggest it will be approximately satisfied in surprisingly many cases. At the very least, assuming simultaneous linearisability is a far-reaching generalisation of assuming multivariate normality.

In all three mathematical examples we used the direct sum of the $\mathbf{K}_j$ to diagonalise the $\mathbf{E}_{j\ell}$, then use a permutation matrix $\mathbf{P}$ to transform $\mathbf{F} = \mathbf{K}'\mathbf{E}\mathbf{K}$ into the direct sum $\mathbf{R} = \mathbf{P}'\mathbf{F}\mathbf{P}$ of correlation matrices, and then use the direct sum $\mathbf{L}$ to diagonalise $\mathbf{R}$ to $\mathbf{\Lambda} = \mathbf{L}'\mathbf{R}\mathbf{L}$. This means that $\mathbf{KPL}$ has the eigenvectors of $\mathbf{E}$, but not ordered by decreasing or increasing eigenvalues. It also means that the eigenvectors have a special structure.

First, $\mathbf{F}$ is an $m \times m$ matrix of matrices $\mathbf{F}_{j\ell}$, which are $k_j \times k_\ell$. If all k_j are equal to, say, k , then $\mathbf{R}$ is a $k \times k$ matrix of matrices $\mathbf{R}_{st}$, which are all of order m . If the variables have a different number of categories, then $\mathbf{R}$ is a $k_+ \times k_+$ matrix of correlation matrices, with $\mathbf{R}_{st}$ of order $m_s \times m_t$, where m_s is defined as before as the number of variables with $k_j \geq s$.

$\mathbf{KP}$ is an orthonormal $m \times k_+$ matrix of matrices, in which column-block s is the direct sum of the m_s column vectors $\mathbf{K}_j \mathbf{e}_s$, with $\mathbf{e}_s$ unit vector s (equal to zero, except for element s , which is one). In a formula $\{\mathbf{KP}\}_{js} = \mathbf{K}_j \mathbf{e}_s \mathbf{e}_s'$ and $\{\mathbf{KP}\}_{js} \mathbf{L}_s = \mathbf{K}_j \mathbf{e}_s \mathbf{e}_s' \mathbf{L}_s$. Matrix $\{\mathbf{KPL}\}_{js}$ is the $k_j \times m_s$ outer product of column s of $\mathbf{K}_j$ and row s of $\mathbf{L}_s$. Each $\mathbf{R}_{ss}$ is computed with a single quantification of the variables, and there are only $k_+ - 1$ different non-trivial quantifications, instead of the $k_\star - m$ ones from MCA.

That the matrix $\mathbf{KPL}$ is blockwise of rank one connects DMCA with non-linear PCA, which is MCA with rank one restrictions on the category quantifications. We see that imposing rank one restrictions on MCA forces non-linear PCA to choose its solutions from the same $\mathbf{R}_{ss}$, thus preventing “data production”.

5 The Chi-Square Metric

In the Correspondence Analysis of a single table it has been known since Hirschfeld (1935) that the sum of squares of the non-trivial singular values is equal to the chi-square (the total inertia) of the table. Although both Burt and Guttman pay homage to chi-square in the context of MCA, they do not really work through the consequences. In this section we analyse the total chi-square (TCS), which is the sum of all $m(m-1)$ off-diagonal bivariate chi-squares.

De Leeuw (1973, p. 32), shows that the TCS is related to the MCA eigenvalues by the simple equation:

$$\sum_{1 \leq j \neq \ell \leq m} \mathcal{X}_{j\ell}^2 = n \sum_s (m\lambda_s - 1)^2, \quad (7)$$

where the sum on the right is over all $k_\star - m$ non-trivial eigenvalues. The same formula was given by Benzécri (1979). Equation (7), the MCA decomposition of the TCS, gives us a way to quantify the contribution of each non-trivial eigenvalue.

We now outline the DMCA decomposition of the TCS. An identity similar to (7) is:

$$\sum_{1 \leq j \neq \ell \leq m} \mathcal{X}_{j\ell}^2 = \text{tr } \mathbf{E}^2 - (K + m(m-1)). \quad (8)$$

Equation (8) does not look particularly attractive, until one realises that the constant subtracted on the right is the number of trivial elements in $\mathbf{F} = \mathbf{K}'\mathbf{E}\mathbf{K}$ (and thus in $\mathbf{R} = \mathbf{P}'\mathbf{K}'\mathbf{E}\mathbf{P}\mathbf{K}$) equal to one. There are K elements on the main diagonal, and $m(m-1)$ elements from the off-diagonal elements of the trivial matrix $\mathbf{R}_{11g}$.

Thus the TCS can be partitioned using $\mathbf{R}$, which is a $k_+ \times k_+$ matrix of matrices into $(k_+ - 1)^2$ non-trivial components. The most interesting ones are the $k_+ - 1$ sums of squares of the off-diagonal elements of the diagonal submatrices $\mathbf{R}_{22}, \dots, \mathbf{R}_{k_+k_+}$, which is actually the quantity maximised by DMCA. And then there are the $(k_+ - 1)(k_+ - 2)$ sums of squares of the off-diagonal submatrices of $\mathbf{R}$, which is

actually what DMCA minimises. The sum of squares of each diagonal block separately is its contribution to the DMCA fit, and total contribution to chi-square over all diagonal blocks shows how close DMCA is to MCA, i.e. how well DMCA diagonalises $\mathbf{E}$. In the mathematical examples from Sect. 4.2 DMCA is just a rearranged MCA, and all of the TCS comes from the diagonal blocks.

6 Computation

So, computationally, DMCA works in three steps. All three steps preserve orthonormality, guaranteeing that if DMCA diagonalisation works we have actually found eigenvalues and eigenvectors of $\mathbf{E}$, i.e. the MCA solution.

In the first step we compute the $\mathbf{K}_j$ by approximately diagonalising all off-diagonal $\mathbf{E}_{j\ell}$. This is done in the mathematical examples by using known analytical results, but in empirical examples by Jacobi rotations that minimise the sum of squares of all off-diagonal elements of the off-diagonal $\mathbf{K}'\mathbf{E}\mathbf{K}$ (or, equivalently, maximise the sum of squares of the diagonal elements).

Each $\mathbf{K}_j$ is $k_j \times k_j$ and square orthonormal. We always set the first column of $\mathbf{K}_j$ equal to $n^{-\frac{1}{2}}\sqrt{\mathbf{d}_j}$, with d_j the marginals of variable j , to make sure the first column captures the non-zero trivial solution. This is done by setting the initial $\mathbf{K}_j$ to the left singular vectors of row-block j of $\mathbf{E}$ and not rotating pairs of indices (s, t) when s or t is one. This usually turns out to be a very good initial solution.

In the second step we permute the rows and columns of $\mathbf{F} = \mathbf{K}'\mathbf{E}\mathbf{K}$ into direct sum form. The $(1, 1)$ matrix $\mathbf{R}_{11}$ in $\mathbf{R} = \mathbf{P}'\mathbf{K}'\mathbf{E}\mathbf{K}\mathbf{P}$ has the $(1, 1)$ elements of all $\mathbf{F}_{j\ell}$, the $(1, 2)$ matrix $\mathbf{R}_{12}$ has the $(1, 2)$ elements of all $\mathbf{F}_{j\ell}$, and so on. Thus, if the first step has diagonalised all off-diagonal $\mathbf{E}_{j\ell}$, then all off-diagonal matrices in $\mathbf{R}$ are zero. The square symmetric matrices along the diagonal, of which there are k_+ , are of order m , or of order m_s if not all k_j are equal. The first two, $\mathbf{R}_{11}$ and $\mathbf{R}_{22}$, are always of order m . $\mathbf{R}_{11}$ takes care of all m trivial solutions and has all its elements equal to one.

Then, in the third step, we diagonalise the matrices along the diagonal of $\mathbf{R}$ by computing their eigenvalues and eigenvectors. This gives $\mathbf{\Lambda} = \mathbf{L}'\mathbf{R}\mathbf{L}$, which is diagonal if the first step succeeded in diagonalising all off-diagonal $\mathbf{E}_{j\ell}$. All the loss that can make DMCA an imperfect diagonalisation method is in the first step, computing both $\mathbf{P}$ and $\mathbf{L}$ does not introduce any additional loss. Note again that the direct sums of $\mathbf{K}$ and $\mathbf{L}$ and the permutation matrix $\mathbf{P}$ are all orthonormal, and thus so are $\mathbf{K}\mathbf{P}$ and $\mathbf{K}\mathbf{P}\mathbf{L}$.

Finally we compute $\mathbf{Y}'\mathbf{K}\mathbf{P}\mathbf{L}$, with $\mathbf{Y}$ the MCA solution, to see how close $\mathbf{Y}$ and $\mathbf{K}\mathbf{P}\mathbf{L}$ are, and which $\mathbf{R}_{ss}$ the MCA solutions come from. Note that $\mathbf{Y}'\mathbf{K}\mathbf{P}\mathbf{L}$ is also square orthonormal, which implies sums of squares of rows and columns add up to one, and squared elements can be interpreted as proportions of “variance explained”.

DMCA has an interesting relationship with the Ordered Multiple Correspondence Analysis (OMCA) of Lombardo and Meulman (2010). DMCA choose the $\mathbf{K}_j$ that

make the E_{jl} as diagonal as possible, in order to concentrate as much of the TCS in the diagonal correlation matrices R_{ss} . In OMCA the K_j are chosen as orthogonal polynomials for variable j of degrees $0, \dots, k_j - 1$, with again K their direct sum. Then compute $F = K'EK$ and $R = P'FP$ and $\Lambda = L'RL$ as in DMCA. This gives the same type of partitioning of the TCS, and the same blockwise rank one approximate eigenvectors KPL , but of course with less of the total TCS concentrated on the diagonal. In the case of binary data and a continuous multinomial OMCA and DMCA are the same. If there are only two variables they are different, and the OMCA results are a rearrangement of those in Beh (1997). Of course if the K_j computed by DMCA are not polynomials, for example if categories are unordered nominal, the two methods can give very different results. But a more detailed comparison on various real examples would be useful. The web directory <https://jansweb.netlify.app/post/code/> also has R code for MCA and OMCA.

6.1 The Programme

For the empirical examples in the present paper we use the R function `DMCA`, a further elaboration of the R function `jMCA` from De Leeuw and Ferrari (2008). The programme, and all the empirical examples with the necessary data manipulations, can be downloaded from <https://jansweb.netlify.app/post/code/>. The programme maximises the percentage of the TCS in the diagonal blocks of the DMCA. It is called with arguments:

- `burt`, the Burt matrix,
- `k`, the number of categories of the variables,
- `eps`, iteration precision, defaults to $1e-8$,
- `itmax`, maximum number of iterations, defaults to 500,
- `verbose`, prints DMCA fit for all iterations, defaults to `TRUE`,
- `vectors`, DMCA eigenvectors, if `FALSE` only DMCA eigenvalues, defaults to `TRUE`,

and it returns a list with

- `kek`, the matrix $K'EK$,
- `pkek`, the matrix $P'K'EK P$,
- `lpkekpl`, the matrix $L'P'K'EK PL$,
- `k`, the block-diagonal matrix K ,
- `p`, the permutation P ,
- `l`, the block-diagonal matrix L ,
- `kp`, the matrix $K P$,
- `kpl`, the matrix $K P L$,
- `chisquares`, the $m(m-1)$ chi-squares
- `chipartition`, the DMCA chi-partition,
- `chippercentages` = `chipartition` / TCS,

- `itel`, the number of iterations,
- `func`, the optimum value of trace of chipercentages

7 Empirical Examples

We analysed DMCA in our previous examples by relying solely on specific mathematical properties. There are some empirical examples in the last section of De Leeuw (1982), but with very little detail, and computed with a now tragically defunct APL programme. Showing the matrices $\mathbf{K}$, $\mathbf{P}$, $\mathbf{L}$ as well as $\mathbf{F}$, $\mathbf{R}$ and $\mathbf{A}$ in this paper would take up too much space, so we concentrate on how well DMCA reproduces the MCA eigenvalues. We also discuss which of the correlation matrices in $\mathbf{R}$ the first and last MCA vectors of weights (eigenvectors) are associated with, and we give the partitionings of the TCS.

7.1 Burt Data

The data for the example in Burt (1950) were collected by him in Liverpool in or before 1912, and are described in an outrageously politically incorrect paper (Burt 1912). Burt used $m = 4$, with variables hair-colour (fair, red, dark), eye colour (light, mixed, brown), head (narrow, wide), and stature (tall, short) for 100 individuals selected from his sample. This is not very interesting as a DMCA or MCA example because the data are so close to binary and thus there is not much room for DMCA to work with. We include the Burt data, using the Burt table from Burt (1950), for historical reasons.

The Burt table is of order $k_* = 10$, so there are $k_* - m = 6$ non-trivial eigenvalues. DMCA takes one single iteration cycle to convergence to fit 0.9462 from the initial SVD solution. Figure 1 plots the sorted MCA and DMCA non-trivial eigenvalues. In these plots we always remove the trivial points (0, 0) and (1, 1) because they would anchor the plot and unduly emphasise the closeness of the two solutions.

The matrix $\mathbf{R}$ has two diagonal blocks, $\mathbf{R}_{11}$ and $\mathbf{R}_{22}$, of order four and one block, $\mathbf{R}_{33}$, of order two. Thus the m_s are (4, 4, 2). The first non-trivial MCA solution correlates 0.9997 with the first non-trivial DMCA solution, which corresponds with the dominant eigenvalue of $\mathbf{R}_{22}$. The second MCA solution correlates -0.7319 with the second DMCA solution from $\mathbf{R}_{22}$ and -0.3749 and -0.5675 with the two DMCA solutions from $\mathbf{R}_{33}$. The fifth and sixth MCA solutions (the ones with the smallest non-trivial eigenvalues) correlate 0.9824 and 0.9937 with the remaining two DMCA solutions from $\mathbf{R}_{22}$. Thus, almost all the variation comes from $\mathbf{R}_{22}$, because with the k_j as small as (3, 3, 2, 2) we are very close to the case where all variables only take two values and all the variation is in the phi-coefficients in $\mathbf{R}_{22}$.

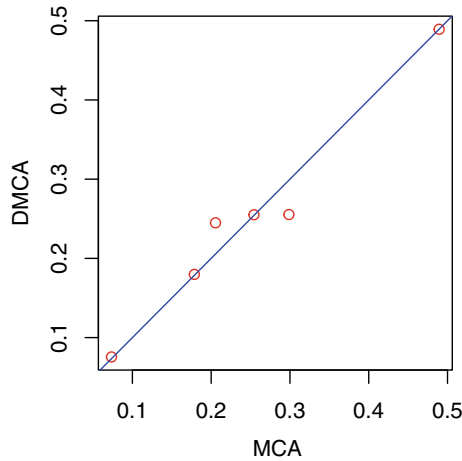


Fig. 1 Burt MCA/DMCA eigenvalues

We can further illustrate this with the chi-square partitioning. Of the TCS of 156.68 the diagonal blocks $\mathbf{R}_{22}$ and $\mathbf{R}_{33}$ contribute, respectively, 148.1664 (95%) and 0.08237 (0.05%), while the off-diagonal blocks contribute 8.4319 (5%).

7.2 GALO Data

The GALO data (Peschar 1975) are a mainstay Gifi example. The individuals are $n = 1290$ sixth grade school children in the city of Groningen, The Netherlands, about to go into secondary education. The $m = 4$ variables are gender (2 categories), IQ (9 categories), teachers advice (7 categories), and socio-economic status (6 categories). The Burt matrix is of order $k_* = 24$, and thus there are $k_* - m = 20$ non-trivial dimensions. Matrix $\mathbf{R} = \mathbf{P}'\mathbf{F}\mathbf{P}$ has 9 diagonal correlation blocks, with $\mathbf{R}_{11}$ and $\mathbf{R}_{22}$ of order four, $\mathbf{R}_{33}, \dots, \mathbf{R}_{66}$ of order three, $\mathbf{R}_{77}$ of order two, and $\mathbf{R}_{88}$ and $\mathbf{R}_{99}$ of order one. DMCA takes 37 iteration cycles to a fit of 0.8689. The 20 sorted non-trivial MCA and DMCA eigenvalues are plotted in Fig. 2.

The strong Guttman effect in the GALO data is reflected in the close correspondence between the MCA and DMCA solutions. The first non-trivial MCA solution correlates 0.9967 with the dominant DMCA solution from $\mathbf{R}_{22}$, and the second MCA solution correlates 0.9915 with the dominant DMCA solution from $\mathbf{R}_{33}$. After that correlations become smaller, until we get to the smallest eigenvalues. The worst MCA solution correlates -0.9882 with the solution corresponding to smallest eigenvalue of $\mathbf{R}_{22}$, and the next worst correlates -0.9794 with the solution with the smallest eigenvalue of $\mathbf{R}_{33}$.

To illustrate graphically how close MCA and DMCA are we plot the 24 category quantifications on the first non-trivial dimension of the MCA solution (MCA dimen-

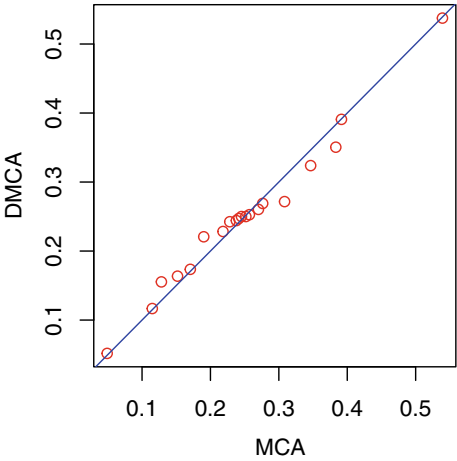


Fig. 2 GALO MCA/DMCA eigenvalues

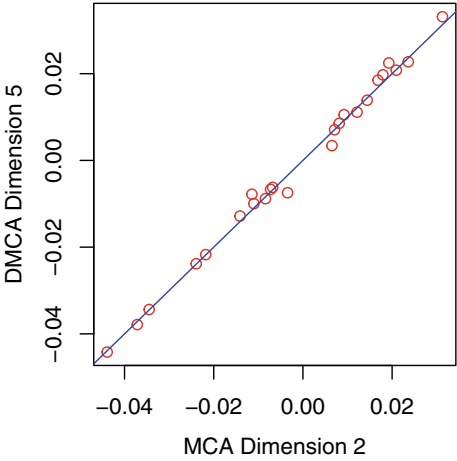


Fig. 3 GALO MCA/DMCA quantifications

sion two) and the first non-trivial dimension of DMCA (dimension five) in Fig. 3. Note the dominant MCA dimension is always the trivial one, so we need the second MCA dimension. For DMCA the first four dimensions correspond with the trivial $\mathbf{R}_{11}$, and thus the first interesting dimension is number $m + 1$, corresponding with the dominant eigenvalue of $\mathbf{R}_{22}$. In Fig. 4 we plot the corresponding MCA dimension three and DMCA dimension $2m + 1 = 9$, corresponding with the dominant eigenvalue of $\mathbf{R}_{33}$.

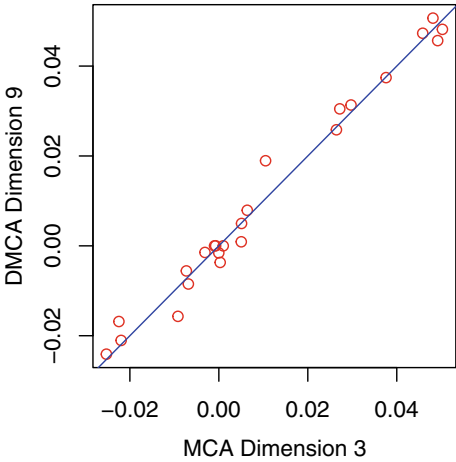


Fig. 4 GALO MCA/DMCA quantifications

Table 1 GALO TCS percentages

	DMCA2	DMCA3	DMCA4	DMCA5	DMCA6	DMCA7	DMCA8	DMCA9
DMCA2	0.5631	0.0020	0.0215	0.0172	0.0146	0.0024	7e-04	1e-04
DMCA3	0.0020	0.1638	0.0003	0.0001	0.0011	0.0001	0e+00	4e-04
DMCA4	0.0215	0.0003	0.0831	0.0003	0.0010	0.0001	0e+00	4e-04
DMCA5	0.0172	0.0001	0.0003	0.0492	0.0016	0.0008	0e+00	1e-04
DMCA6	0.0146	0.0011	0.0010	0.0016	0.0058	0.0001	5e-04	0e+00
DMCA7	0.0024	0.0001	0.0001	0.0008	0.0001	0.0041	0e+00	0e+00
DMCA8	0.0007	0.0000	0.0000	0.0000	0.0005	0.0000	0e+00	0e+00
DMCA9	0.0001	0.0004	0.0004	0.0001	0.0000	0.0000	0e+00	0e+00

The chi-square partitioning tells us the diagonal blocks of DMCA “explain” 87% of the TCS, with the blocks $\mathbf{R}_{22}$, $\dots$, $\mathbf{R}_{77}$ contributing 56, 16, 8, 5, 0.5, and 0.4%. The complete partitioning is summarised in Table 1.

7.3 BFI Data

Our final example is larger, and somewhat closer to an actual application of MCA. The BFI data set is taken from the `psychTools` package (Revelle 2021). It has $n = 2800$ observations on $m = 25$ personality self report items. After removing persons with missing data there are $n = 2436$ observations left. Each item has $k = 6$ categories, and thus the Burt table is of order $m \times k = 150$. Matrix $\mathbf{R}$, excluding $\mathbf{R}_{11}$, has five diagonal blocks of order 25. DMCA takes 54 iterations for a DMCA

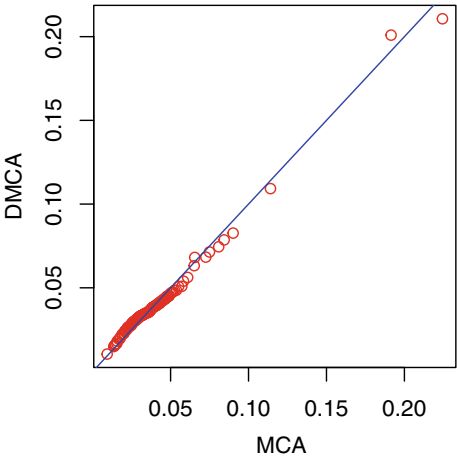


Fig. 5 BFI MCA/DMCA eigenvalues

Table 2 BFI TCS percentages

	DMCA2	DMCA3	DMCA4	DMCA5	DMCA6
DMCA2	0.4877	0.0153	0.0059	0.0055	0.0041
DMCA3	0.0153	0.3302	0.0053	0.0037	0.0035
DMCA4	0.0059	0.0053	0.0394	0.0049	0.0042
DMCA5	0.0055	0.0037	0.0049	0.0206	0.0046
DMCA6	0.0041	0.0035	0.0042	0.0046	0.0081

fit of 0.8860. The sorted non-trivial 125 MCA and DMCA eigenvalues are plotted in Fig. 5.

The percentages of the TCS from the non-trivial submatrices of $\mathbf{R}$ are summarised in Table 2.

8 Discussion

Our mathematical and empirical examples show that in a wide variety of circumstances MCA and DMCA eigenvalues and eigenvectors are very similar, although DMCA uses far fewer degrees of freedom for its diagonalisation. This indicates that DMCA can be thought of, at least in some circumstances, as a smooth version of MCA. The error is moved to the off-diagonal elements in the submatrices of $\mathbf{R} = \mathbf{P}^t\mathbf{F}\mathbf{P}$ and the structure is concentrated in the diagonal correlation matrices.

We have also seen that DMCA is like MCA, in the sense that it gives very similar solutions, but it is also like non-linear PCA, because it imposes the rank one restric-

tions on the weights. Thus it is a bridge between the two techniques, and it clarifies their relationship.

DMCA also shows where the dominant MCA solutions originate, and indicates quite clearly where the Guttman effect comes from (if it is there). It suggests the Guttman effect, in a generalised sense, does not necessarily result in polynomials or arcs. As long as there is simultaneous linearisation of all bivariate regressions $\mathbf{E}$ is orthonormally similar to the direct sum of the $\mathbf{R}_{ss}$, and the principal components of the $\mathbf{R}_{ss}$ will give a generalised Guttman effect.

This allows us to suggest some answer for questions coming from the Burt-Guttman exchange. In many cases the principal components of MCA (beyond the first) come from the generalised Guttman effect, and should be interpreted as such. Thus the first principal component does have a special status, and thus justifies singling out RAA and Guttman scaling from the rest of MCA.

DMCA also reduces the amount of data production. Instead of $k_* - m$ non-trivial correlations matrices of order m with their PCA's, we now have $k_+ - 1$ non-trivial correlation matrices of orders given by the m_s . That is still more than one single correlation matrix, as we have in non-linear PCA and the aspect approach, but the different correlation matrices may either be related by the Guttman effect or give non-trivial additional information.

We also mention some other attempts, besides (7) and DMCA, to deal with the influence of the diagonal blocks on the MCA solution. The first is Greenacre's Joint Correspondence Analysis or JMCA (Greenacre 1988), which minimises $\mathbf{E} - \mathbf{U}\mathbf{U}'$ not only over all $K \times p$ matrices $\mathbf{U}$ with $\mathbf{U}'\mathbf{U} = \mathbf{I}$, but in addition over the m diagonal blocks of $\mathbf{E}$. In JMCA the dominant trivial dimension is first removed. JMCA uses a variation of the Thomson's alternating least squares algorithm for least squares factor analysis, alternating the minimising over $\mathbf{U}$ for given $\mathbf{C}$ and the minimising over the diagonal blocks of $\mathbf{C}$ for given $\mathbf{U}$. The first minimisation is an MCA of the modified Burt matrix with the current diagonal blocks, the second minimisation replaces the diagonal blocks of $\mathbf{C}$ with the corresponding ones of $\mathbf{U}\mathbf{U}'$. As a result JMCA does optimise the fit to the TCS without the adjustments of (7). Nevertheless there are some problems with JMCA. It fixes the dimension p at a low value, and can compute separate un-nested solutions for each p . Thus it tends to "data production" in our sense, because we have to find a way to relate the solutions for different p . As in DMCA and MCA it would be advantageous to have a complete and simultaneous nested solution by always choosing $p = K - m$. The second problem with JMCA is that, when p becomes larger, Heywood cases may become more common, i.e. cases in which the reduced Burt matrix is no longer positive semi-definite. This potentially leads to complex numbers and negative variances.

The second way of dealing with the undesirable dimensionality and explained variances aspects of MCA is not to require $\mathbf{U}'\mathbf{U} = \mathbf{I}$ but $\mathbf{U}'_j\mathbf{U}_j = \mathbf{I}$ for all j . This is sometimes called strong orthogonality (Dauxois and Pousse 1976). We could call the resulting technique strong multiple correspondence analysis of SMCA. If $m = 2$ SMCA still gives MCA, and thus also CA and JMCA, but if $m > 2$ SMCA is only MCA or JMCA if we have simultaneous linearisability. SMCA tends to make all variables equally important; see the discussion in Nishisato and Sheu (1980). SMCA

also has its problems. The constraint $\mathbf{U}_j' \mathbf{U}_j = \mathbf{I}$ limits the dimensionality of the non-trivial quantifications for variable j to $k_j - 1$, and it is unclear what to do with the higher dimensions in $\mathbf{E}$. In DMCA strong orthogonality constraints are imposed on the $\mathbf{K}_j$, but the columns of the $\mathbf{K}_j$ are distributed over different correlation matrices, and the resulting $\mathbf{U}_j$ are of rank one, but no longer orthonormal. The mathematical properties of both JMCA and SMCA deserve some further study.

This also seems the place to point out a neglected aspect of MCA. The smallest non-trivial solution gives a quantification or transformation of the data that maximises the singularity of the transformed data, i.e. the minimum eigenvalue of the corresponding correlation matrix. We have seen in our empirical examples that MCA and DMCA often agree closely in their smallest eigenvalue solutions, and that may indicate that it should be possible to give a scientific interpretation of these “bad” solutions. In fact, the smallest DMCA and MCA eigenvalues can be used in a regression interpretation in which we consider one or more of the variables as criteria and the others are predictors.

A complaint that many users of MCA have is that, say, the first two components “explain” such a small proportion of the “variance”, by which they mean the trace of $\mathbf{E}$, which is $\mathbf{K}$, the total number of categories, and which, of course, has nothing to do with “variance”. Equation (7) indicates how to quantify the contributions of the non-trivial eigenvalues. For the BFI data, for example, the first two non-trivial MCA eigenvalues “explain” 0.0832 percent of the “variance”, but they “explain” 0.6305 percent of the TCS. Moreover DMCA shows us that we should really relate the eigenvalues to the $\mathbf{R}_{ss}$ they come from, and see how much they “explain” of their correlation matrices. It is even better to evaluate their contributions using the TCS and its partitioning described in Sect. 5 of this paper.

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